
CHAPTER 1

UNITS, PHYSICAL QUANTITIES AND VECTORS

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Intended Learning Outcomes

At the end of this chapter, the students must be able to:

- **Identify** physical quantities and their SI/Imperial units.
- **Convert** accurately between different units of measurement.
- **Distinguish** between scalar and vector quantities, providing practical examples.
- **Represent** vector quantities graphically (e.g., forces, displacements).
- **Apply** graphical methods (e.g., parallelogram, polygon) and component method for vector addition and subtraction to solve for the resultant vector.

1 What Physics Is

Think about the technological devices you use daily: computers, smartphones, wireless internet, GPS, and countless others. Consider also the groundbreaking advancements you hear about, like levitating trains or microscopic robots. All of these innovations, and indeed much of the world around us, rely on the fundamental principles of **physics**.

Physics is a branch of science, derived from a Latin word meaning ‘having knowledge.’ It refers to the knowledge of how the physical world operates, based on objective evidence determined through observation and experimentation. A key requirement of any scientific explanation is that it must be **testable** through investigation.

At its core, physics is the science aimed at describing the fundamental aspects of our universe. This includes what things are in it, their noticeable properties, and the processes they undergo. Simply put, physics attempts to describe the basic mechanisms that make our universe behave the way it does. For instance, physics explains how electric current interacts with circuits inside a smartphone, guiding engineers in material selection and circuit layout. Similarly, for a GPS device, physics describes the relationship between an object’s speed, distance, and time, allowing it to determine travel times between locations.

Over centuries, as technology evolved, physics expanded into many branches. Early studies focused on **kinematics** (the study of moving objects), **statics**, and **dynamics** (how objects respond to forces). This foundational knowledge led to the invention of simple machines like levers and pulleys, which enabled humans to do more work effectively.

Later, branches like **acoustics** (the study of sound) and **optics** (the study of light) emerged. The invention of the telescope, for example, revolutionized astronomy. Isaac Newton’s laws of motion, built upon Galileo’s observations, became the standard for kinematics and dynamics. **Thermodynamics**, the study of thermal energy and heat transfer, was spurred by the need for more efficient engines and food preservation.

The 18th and 19th centuries saw significant advancements in **electricity** and **magnetism**. Discoveries by Oersted and Faraday demonstrated the intrinsic link between electric currents and magnetic fields, leading directly to inventions like the electric motor and generator that revolutionized human life.

More recently, the 20th century introduced **nuclear physics** (studying atomic nuclei) and **quantum mechanics** (dealing with atoms and subatomic particles), leading to technologies like nuclear power. Albert Einstein’s **theory of relativity** transformed our understanding of motion, space, and time, particularly at very high speeds or in strong gravitational fields. These modern theories, while often counter-intuitive to everyday experience, are crucial for understanding the universe’s most extreme conditions and form the basis of advanced technologies like GPS satellites, which must account for relativistic effects on time.

In essence, physics provides the analytical abilities to understand the world and solve problems across countless disciplines. It is the foundation upon which chemistry, engineering (including architectural, electrical, mechanical, and food technology), geology, and even medical diagnostics and biology are built. By studying the basic laws of physics, you will significantly enhance your problem-solving skills and critical thinking, which are invaluable in any career path you choose.

2 Physical Quantities and Units

Definition of Physical Quantities

Definition 2.1: Physical Quantity

A **physical quantity** is a property of a phenomenon, body, or substance that can be quantified by measurement.

In simpler terms, if you can measure it, it's a physical quantity. For instance, the length of a component, the mass of an ingredient, or the time it takes for a process to complete are all physical quantities. These quantities provide the framework for describing and understanding the physical world.

Units of Measurement

To express a physical quantity, we need a unit.

Definition 2.2: Unit of Measurement

A **unit of measurement** is a standard quantity used to express a physical quantity.

There are two primary systems of units you will encounter:

- **International System of Units (SI Units):** This is the most widely used system globally, especially in science, engineering, and international commerce. It is based on seven base units (e.g., meter for length, kilogram for mass, second for time).
- **Imperial System:** Also known as the British Imperial System, this system is primarily used in the United States and a few other countries. Common units include feet for length, pounds for mass, and gallons for volume.

Understanding both systems is crucial, as you might encounter specifications or equipment from different parts of the world.

Fundamental vs. Derived Quantities

Physical quantities are categorized into two main types:

- **Fundamental Quantities (Base Quantities):** These are a set of independent physical quantities from which all other physical quantities can be derived. They are considered the building blocks of measurement. The International System of Units (SI) is based on seven fundamental quantities.
- **Derived Quantities:** These quantities are expressed as combinations of fundamental quantities through mathematical relationships (e.g., multiplication, division). For example, *speed* is a derived quantity (length/time), and *area* is a derived quantity (length

$\times$ length). *Pressure*, which is force per unit area, is also a derived quantity, as force and area are themselves derived from fundamental quantities.

Common Fundamental Physical Quantities and Their Units

Here are the seven fundamental physical quantities and their SI units:

Physical Quantity	SI Unit (Symbol)	Common Imperial Unit (Symbol)
Length	meter (m)	foot (ft), inch (in)
Mass	kilogram (kg)	pound (lb)
Time	second (s)	second (s)
Electric Current	ampere (A)	—
Temperature	Kelvin (K)	Fahrenheit ($^{\circ}\text{F}$), Celsius ($^{\circ}\text{C}$)
Amount of Substance	mole (mol)	—
Luminous Intensity	candela (cd)	—

Practice Task 1

For each physical quantity listed below, identify whether it is a **Fundamental Quantity** or a **Derived Quantity**. If it is a Derived Quantity, list the fundamental quantities from which it is derived.

1. **Volume** (e.g., a tank's capacity in cubic meters)
2. **Force** (e.g., the push from a hydraulic cylinder in Newtons)
3. **Time** (e.g., the duration of a manufacturing cycle in seconds)
4. **Density** (e.g., the density of a material in kilograms per cubic meter)
5. **Electric Charge** (e.g., in Coulombs)
6. **Area** (e.g., the surface area of a component in square centimeters)
7. **Velocity** (e.g., the speed and direction of a robotic arm in meters per second)
8. **Temperature** (e.g., the heat of an oven in degrees Celsius)

3 Unit Conversion

Importance of Conversion

In various industrial technology fields, you will often encounter situations where measurements are given in different unit systems or different magnitudes within the same system. For example, a blueprint might be in millimeters, but your tools measure in inches, or a material specification might be in kilopascals while your gauge reads in PSI. Accurate **unit conversion** is essential to ensure precision, compatibility, and safety in all your work. Mistakes in conversion can lead to significant errors in design, manufacturing, and operational processes.

Conversion Techniques

The most reliable way to convert units is using **conversion factors**.

Definition 3.1: Conversion Factor

A **conversion factor** is a ratio that relates two different units and is equal to one.

For example, since 1 inch = 2.54 centimeters, the conversion factor can be written as $\frac{2.54 \text{ cm}}{1 \text{ in}}$ or $\frac{1 \text{ in}}{2.54 \text{ cm}}$. You multiply the quantity you want to convert by the conversion factor that will cancel out the original unit and leave you with the desired unit.

Here is a table of some commonly used conversion factors:

Quantity	Conversion Factor
Length	1 inch = 2.54 cm 1 foot = 0.3048 m 1 mile = 1.609 km
Mass	1 kg = 2.2046 lbs 1 lb = 0.4536 kg
Volume	1 gallon (US) = 3.785 L 1 ft^3 = 28.317 L
Force	1 N = 0.2248 lbs 1 lb = 4.448 N
Pressure	1 psi = 6894.76 Pa 1 atm = 101325 Pa
Temperature	$^{\circ}C = (^{\circ}F - 32) \times \frac{5}{9}$ $^{\circ}F = (^{\circ}C \times \frac{9}{5}) + 32$ $K = ^{\circ}C + 273.15$

Metric Prefixes (The "Moving Decimal" Trick) The SI system uses prefixes to denote multiples or submultiples of base units. This makes conversions within the SI system very straightforward, often just requiring a shift of the decimal point.

Prefix	Symbol	Factor
Giga	G	10^9
Mega	M	10^6
Kilo	k	10^3
Hecto	h	10^2
Deca	da	10^1
—	(base unit)	10^0 (1)
Deci	d	10^{-1}
Centi	c	10^{-2}
Milli	m	10^{-3}
Micro	μ	10^{-6}
Nano	n	10^{-9}

To convert between metric units, simply move the decimal point according to the difference in the exponents of the prefixes.

- Moving from a larger unit to a smaller unit (e.g., kilometers to meters), move the decimal to the *right*.
- Moving from a smaller unit to a larger unit (e.g., millimeters to meters), move the decimal to the *left*.

The number of places to move the decimal is equal to the difference in the exponents of the prefixes.

Illustrative Example :

Convert 2.5 kilometers (km) to meters (m).

Kilo (10^3) to base unit (10^0) is a difference of 3 places. Moving from larger to smaller, shift decimal right.

$$2.5 \text{ km} = 2500 \text{ m}$$

Illustrative Example :

Convert 1500 millimeters (mm) to meters (m).

Milli (10^{-3}) to base unit (10^0) is a difference of 3 places. Moving from smaller to larger, shift decimal left.

$$1500 \text{ mm} = 1.5 \text{ m}$$

Examples of Unit Conversion

1. Fundamental Quantity Conversions

a. SI to SI Conversion (using prefixes) Convert 750 grams (g) to kilograms (kg).

$$750 \text{ g} \times \frac{1 \text{ kg}}{1000 \text{ g}} = 0.75 \text{ kg}$$

b. Imperial to Imperial Conversion Convert 4.5 feet (ft) to inches (in).

$$4.5 \text{ ft} \times \frac{12 \text{ in}}{1 \text{ ft}} = 54 \text{ in}$$

c. SI to Imperial Conversion A component has a length of 20 centimeters (cm). Convert this to inches.

$$20 \text{ cm} \times \frac{1 \text{ in}}{2.54 \text{ cm}} \approx 7.87 \text{ in}$$

d. Imperial to SI Conversion A mass is measured as 10 pounds (lb). Convert this to kilograms (kg).

$$10 \text{ lb} \times \frac{0.4536 \text{ kg}}{1 \text{ lb}} \approx 4.536 \text{ kg}$$

2. Derived Quantity Conversions

a. SI to SI Conversion Convert a speed of 36 kilometers per hour (km/h) to meters per second (m/s).

$$36 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 10 \frac{\text{m}}{\text{s}}$$

b. Imperial to Imperial Conversion Convert a flow rate of 120 gallons per minute (gpm) to cubic feet per minute (ft^3/min). (Given: $1\text{ }ft^3 = 7.48\text{ gallons}$)

$$120 \frac{\text{gallons}}{\text{min}} \times \frac{1\text{ }ft^3}{7.48\text{ gallons}} \approx 16.04 \frac{ft^3}{\text{min}}$$

c. SI to Imperial Conversion Convert a pressure of 500 Pascals (Pa) to pounds per square inch (psi).

$$500\text{ Pa} \times \frac{1\text{ psi}}{6894.76\text{ Pa}} \approx 0.0725\text{ psi}$$

d. Imperial to SI Conversion Convert a density of 62.4 pounds per cubic foot (lb/ft^3) to kilograms per cubic meter (kg/m^3). (Given: $1\text{ lb} = 0.4536\text{ kg}$, $1\text{ ft} = 0.3048\text{ m}$)

$$\begin{aligned} 62.4 \frac{lb}{ft^3} &\times \frac{0.4536\text{ kg}}{1\text{ lb}} \times \left(\frac{1\text{ ft}}{0.3048\text{ m}} \right)^3 \\ &= 62.4 \times 0.4536 \times \frac{1}{(0.3048)^3} \frac{kg}{m^3} \\ &\approx 62.4 \times 0.4536 \times 35.3147 \frac{kg}{m^3} \approx 1000 \frac{kg}{m^3} \end{aligned}$$

Practice Task 2

Solve the following unit conversion problems. Show your work clearly, including the conversion factors used.

1. A structural beam needs to be 3.5 meters long. If you are working with tools that measure in feet and inches, what is this length in feet? ($1\text{ m} = 3.281\text{ ft}$)
2. A food processing machine operates at a temperature of 180 degrees Fahrenheit ($^{\circ}\text{F}$). What is this temperature in degrees Celsius ($^{\circ}\text{C}$)?
3. A conveyor belt moves at a speed of 0.8 meters per second (m/s). What is this speed in kilometers per hour (km/h)?
4. A liquid adhesive has a viscosity measured at 250 centipoise (cP). Convert this to Pascal-seconds ($\text{Pa}\cdot\text{s}$), knowing that $1\text{ cP} = 0.001\text{ Pa}\cdot\text{s}$.
5. An architectural drawing specifies a room area of 40 square meters (m^2). Convert this area to square feet (ft^2).
6. A pipe has an internal diameter of 15 millimeters (mm). What is this diameter in inches?
7. A factory's power consumption is recorded as 5,000 kilowatts (kW) in an hour. Convert this energy consumption to joules (J). (Hint: Power is energy per unit time, $1\text{ W} = 1\text{ J/s}$)
8. A pressure sensor reads 1.5 atmospheres (atm). Convert this pressure to pounds per square inch (psi). ($1\text{ atm} \approx 14.696\text{ psi}$)

4 Introduction to Vectors

Scalars vs. Vectors

Not all physical quantities can be described by just a number and a unit. Some quantities also require a direction.

Definition 4.1: Scalar Quantity

A **scalar quantity** is physical quantity that has only **magnitude** (size or amount). It does not have a direction.

Examples: Temperature (25°C), Mass (5kg), Time (10s), Speed (60km/h), Volume (2L).

Definition 4.2: Vector Quantities

A **vector quantity** is a physical quantity that has both **magnitude** and **direction**.

Examples: Displacement (5m East), Velocity (60km/h North), Force (100N downward), Acceleration (9.8m/s^2 downward), Momentum.

Understanding the difference between scalars and vectors is crucial because they behave differently in calculations and have distinct implications in practical scenarios. For example, knowing the speed of a forklift is useful, but knowing its velocity (speed and direction) is essential for predicting where it will be and preventing collisions.

Vector Representation

Vectors are commonly represented graphically by an **arrow**.

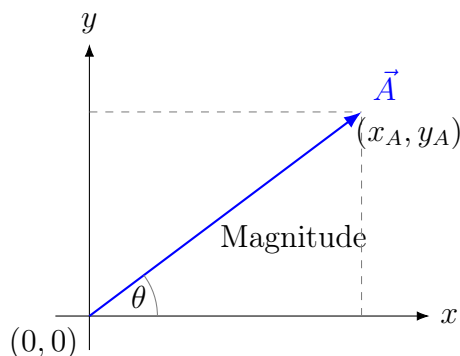


Figure 1.1: Graphical representation of a vector ($\vec{A}$). The arrow's length indicates its magnitude, and the angle θ (or its orientation) indicates its direction. The tail of the vector is at the origin, and the head points to its endpoint (x_A, y_A) .

- The **length** of the arrow represents the **magnitude** of the vector. A longer arrow means a greater magnitude.
- The **direction** in which the arrow points represents the **direction** of the vector.
- The **tail** of the arrow is the starting point, and the **head** (or tip) is the endpoint.

Illustrative Example :

A force of 50 Newtons (N) acting to the right could be represented by an arrow 5 units long pointing right. A force of 100 N to the right would be an arrow 10 units long pointing right.

Understanding Scale in Vector Diagrams: When drawing vectors, we often need to represent very large or very small magnitudes on a manageable diagram. To do this, we use a **scale**. For example, if a force of 50 Newtons (N) acting to the right is represented by an arrow 5 units long (where "units" refers to graphical units on your paper or screen), this implies a scale where 1 graphical unit represents 10 Newtons ($50 \text{ N} / 5 \text{ units} = 10 \text{ N/unit}$). Following this same scale, a force of 100 N to the right would then be represented by an arrow 10 units long ($100 \text{ N} / 10 \text{ N/unit} = 10 \text{ units}$). This scaling allows for accurate visual comparison of vector magnitudes.

Vectors can also be represented algebraically. For instance, a force $\vec{F}$ can be written in terms of its components along axes, like $\vec{F} = F_x \hat{i} + F_y \hat{j}$ in two dimensions, where F_x and F_y are the magnitudes of the components along the x and y axes, and $\hat{i}$ and $\hat{j}$ are unit vectors.

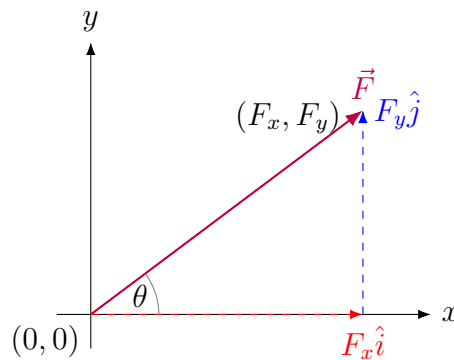


Figure 1.2: Algebraic representation of a vector ($\vec{F}$) using its components. The vector $\vec{F}$ is shown as the sum of its horizontal component ($F_x \hat{i}$) and vertical component ($F_y \hat{j}$).

5 Vector Operations

Why Calculate the Resultant Vector?

In industrial technology, understanding how multiple forces or movements combine is crucial for design, analysis, and troubleshooting. Whether you're dealing with structural integrity, machine dynamics, or fluid flow, objects rarely experience just one force or one movement.

- **Structural Design:** When designing a truss or a support beam, different loads (e.g., weight of components, wind pressure) act at various angles. Calculating the **resultant force** helps engineers determine the total stress on a structure and ensure it won't fail.
- **Machine Dynamics:** A robotic arm might have motors exerting forces in different directions, or a part on a conveyor belt might be pushed by multiple mechanisms. Finding the **resultant velocity** or **resultant acceleration** helps predict the actual path and speed of the object.
- **Fluid Systems:** In piping networks, fluid flow can be influenced by pumps and valves. While more complex, the principles of vector addition can extend to understanding the **resultant flow** in certain scenarios.

Definition 5.1: Resultant Vector

The **resultant vector** represents the single, equivalent vector that has the same effect as all the individual vectors combined. It simplifies complex scenarios into a single, understandable outcome.

Graphical Vector Addition (Finding the Resultant)

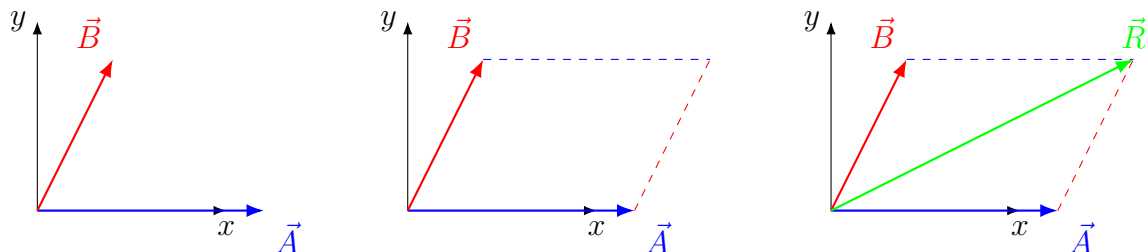
Graphical methods are powerful visual tools for understanding how vectors combine. They allow you to draw vectors to scale and determine the resultant by completing geometric shapes.

Parallelogram Method (For Two Vectors from the Same Point)

This method is ideal when you have two vectors originating from the same point, such as two forces acting on a single object.

Steps:

1. **Draw the Vectors from a Common Origin:** Draw both vector $\vec{A}$ and vector $\vec{B}$ to scale, with their tails starting at the same point.
2. **Complete the Parallelogram:** From the head of $\vec{A}$, draw a dashed line parallel to $\vec{B}$. From the head of $\vec{B}$, draw a dashed line parallel to $\vec{A}$. These dashed lines should intersect, forming a parallelogram.
3. **Draw the Resultant Vector:** The resultant vector, $\vec{R}$, is the diagonal drawn from the common origin to the opposite vertex of the parallelogram.

Illustration - Parallelogram Method:

Step 1: Draw $\vec{A}$ and $\vec{B}$ **Step 2:** Complete Parallelogram **Step 3:** Draw Resultant $\vec{R}$

Figure 1.3: Step-by-step graphical addition using the Parallelogram Method for vectors $\vec{A}$ and $\vec{B}$.

Illustrative Example : (Parallelogram Method)

Two workers are pulling a heavy machine component. Worker 1 pulls with a force of 50 N horizontally to the right ($\vec{F}_1$). Worker 2 pulls with a force of 30 N upwards at a 90° angle to the horizontal ($\vec{F}_2$). Find the resultant force graphically.

Solution:

1. Choose a scale, e.g., 1 cm = 10 N.
2. Draw $\vec{F}_1$ as a 5 cm arrow to the right from the origin.
3. Draw $\vec{F}_2$ as a 3 cm arrow upwards from the same origin.
4. Complete the rectangle (a type of parallelogram) by drawing dashed lines from the heads of $\vec{F}_1$ and $\vec{F}_2$.
5. Draw the diagonal from the origin to the opposite corner. This is $\vec{R}$.
6. Measure the length of $\vec{R}$ and its angle with the horizontal. (You should find $\vec{R} \approx 5.8$ cm (or 58 N) at an angle of approximately 31° above the horizontal). ■

Polygon Method (Head-to-Tail Method - For Two or More Vectors)

This method is versatile and can be used to add any number of vectors.

Steps:

1. **Draw the First Vector:** Draw the first vector ($\vec{A}$) to scale, starting from any convenient point.
2. **Draw Subsequent Vectors Head-to-Tail:** From the head (tip) of the first vector, draw the tail of the second vector ($\vec{B}$) to scale. Continue this process for all subsequent vectors, drawing the tail of each new vector from the head of the previous one.
3. **Draw the Resultant Vector:** The resultant vector ($\vec{R}$) is drawn from the tail of the very first vector to the head of the very last vector.

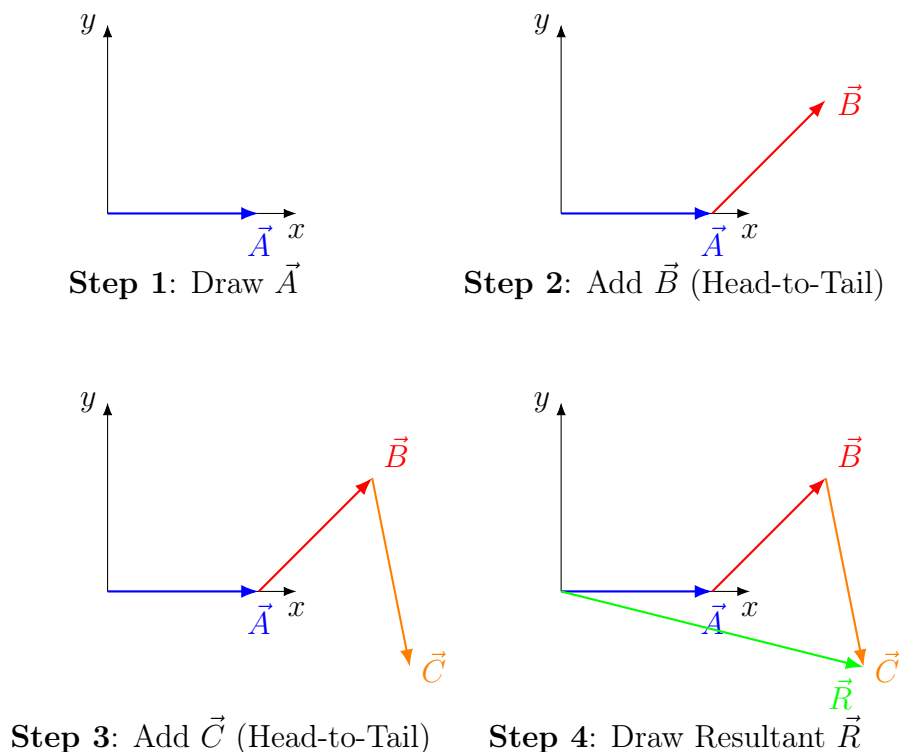
Illustration - Polygon Method:

Figure 1.4: Step-by-step graphical addition using the Polygon (Head-to-Tail) Method for vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$.

Illustrative Example : Polygon Method

A robotic arm makes three consecutive movements:

- Movement 1 ($\vec{D}_1$): 4 m East
- Movement 2 ($\vec{D}_2$): 3 m at 45° North of East
- Movement 3 ($\vec{D}_3$): 2 m South

Find the total displacement (resultant vector) of the robotic arm graphically.

Solution:

1. Choose a scale, e.g., 1 cm = 1 m.
2. Draw $\vec{D}_1$ as a 4 cm arrow horizontally to the East.
3. From the head of $\vec{D}_1$, draw $\vec{D}_2$ as a 3 cm arrow at 45° North of East.
4. From the head of $\vec{D}_2$, draw $\vec{D}_3$ as a 2 cm arrow straight South (downwards).
5. Draw the resultant vector $\vec{R}$ from the starting point (tail of $\vec{D}_1$) to the final point (head of $\vec{D}_3$).
6. Measure the length of $\vec{R}$ (magnitude) and its angle from the East axis (direction). ■

Component Method for Vector Addition

While graphical methods are great for visualization, the **component method** offers a more precise and analytical approach to adding vectors, especially when high accuracy is required or when dealing with many vectors. This method involves breaking down each vector into its perpendicular components (usually along the x and y axes), adding the respective components, and then recombining them to find the resultant vector's magnitude and direction.

Steps:

1. **Resolve Each Vector into Components:** For each vector ($\vec{A}$, $\vec{B}$, etc.), determine its x-component (A_x , B_x) and y-component (A_y , B_y).

- If a vector $\vec{V}$ has magnitude V and makes an angle θ with the positive x-axis:

$$V_x = V \cos(\theta)$$

$$V_y = V \sin(\theta)$$

Remember to pay attention to the quadrant of the vector to get the correct signs for the components.

2. **Sum the Components:** Add all the x-components together to get the resultant x-component (R_x), and add all the y-components together to get the resultant y-component (R_y).

$$R_x = A_x + B_x + \dots$$

$$R_y = A_y + B_y + \dots$$

3. **Calculate the Magnitude of the Resultant Vector:** Use the Pythagorean theorem with the resultant components.

$$R = \sqrt{R_x^2 + R_y^2}$$

4. **Calculate the Direction of the Resultant Vector:** Use the inverse tangent function.

$$\theta_R = \arctan\left(\frac{R_y}{R_x}\right)$$

Important Note: The arctan function typically gives an angle between -90° and $+90^\circ$. You may need to adjust the angle based on the signs of R_x and R_y to place the resultant vector in the correct quadrant (e.g., if R_x is negative and R_y is positive, the vector is in the second quadrant).

Illustrative Example : Component Method

A mounting bracket is subjected to two forces:

- $\vec{F}_1$: 100 N at 30° above the positive x-axis.
- $\vec{F}_2$: 150 N at 120° from the positive x-axis (i.e., 60° above the negative x-axis).

Find the magnitude and direction of the resultant force.

Solution: Step 1: Resolve Each Vector into ComponentsFor $\vec{F}_1$:

$$F_{1x} = 100 \cos(30^\circ) = 100 \times 0.866 = 86.6 \text{ N}$$

$$F_{1y} = 100 \sin(30^\circ) = 100 \times 0.500 = 50.0 \text{ N}$$

For $\vec{F}_2$:

$$F_{2x} = 150 \cos(120^\circ) = 150 \times (-0.500) = -75.0 \text{ N}$$

$$F_{2y} = 150 \sin(120^\circ) = 150 \times 0.866 = 129.9 \text{ N}$$

Step 2: Sum the Components

$$R_x = F_{1x} + F_{2x} = 86.6 \text{ N} + (-75.0 \text{ N}) = 11.6 \text{ N}$$

$$R_y = F_{1y} + F_{2y} = 50.0 \text{ N} + 129.9 \text{ N} = 179.9 \text{ N}$$

Step 3: Calculate the Magnitude of the Resultant Vector

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(11.6)^2 + (179.9)^2}$$

$$R = \sqrt{134.56 + 32364.01} = \sqrt{32498.57} \approx 180.3 \text{ N}$$

Step 4: Calculate the Direction of the Resultant Vector

$$\theta_R = \arctan\left(\frac{R_y}{R_x}\right) = \arctan\left(\frac{179.9}{11.6}\right)$$

$$\theta_R \approx \arctan(15.5086) \approx 86.3^\circ$$

Since both R_x and R_y are positive, the resultant vector is in the first quadrant, so the angle is 86.3° from the positive x-axis.

Result: The resultant force is approximately 180.3 N at 86.3° from the positive x-axis. ■

Illustrative Example : Component Method

A material handler moves in two stages:

- $\vec{D}_1$: 20 m at 0° (due East).
- $\vec{D}_2$: 30 m at 270° (due South).
- $\vec{D}_3$: 15 m at 225° (South-West).

Find the total displacement (magnitude and direction) of the material handler using the component method.

Solution: Step 1: Resolve Each Vector into ComponentsFor $\vec{D}_1$:

$$D_{1x} = 20 \cos(0^\circ) = 20 \times 1 = 20 \text{ m}$$

$$D_{1y} = 20 \sin(0^\circ) = 20 \times 0 = 0 \text{ m}$$

For $\vec{D}_2$:

$$D_{2x} = 30 \cos(270^\circ) = 30 \times 0 = 0 \text{ m}$$

$$D_{2y} = 30 \sin(270^\circ) = 30 \times (-1) = -30 \text{ m}$$

For $\vec{D}_3$:

$$D_{3x} = 15 \cos(225^\circ) = 15 \times (-0.707) = -10.61 \text{ m}$$

$$D_{3y} = 15 \sin(225^\circ) = 15 \times (-0.707) = -10.61 \text{ m}$$

Step 2: Sum the Components

$$R_x = D_{1x} + D_{2x} + D_{3x} = 20 \text{ m} + 0 \text{ m} + (-10.61 \text{ m}) = 9.39 \text{ m}$$

$$R_y = D_{1y} + D_{2y} + D_{3y} = 0 \text{ m} + (-30 \text{ m}) + (-10.61 \text{ m}) = -40.61 \text{ m}$$

Step 3: Calculate the Magnitude of the Resultant Vector

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(9.39)^2 + (-40.61)^2}$$

$$R = \sqrt{88.1721 + 1649.1721} = \sqrt{1737.3442} \approx 41.68 \text{ m}$$

Step 4: Calculate the Direction of the Resultant Vector

$$\theta_R = \arctan\left(\frac{R_y}{R_x}\right) = \arctan\left(\frac{-40.61}{9.39}\right)$$

$$\theta_R \approx \arctan(-4.3248) \approx -76.99^\circ$$

Since R_x is positive and R_y is negative, the resultant vector is in the fourth quadrant. The angle from the positive x-axis (East) is approximately $360^\circ - 76.99^\circ = 283.01^\circ$. Or, more commonly, 76.99° South of East.

Result: The total displacement of the material handler is approximately 41.68 m at an angle of 283.01° (or 76.99° South of East) from the positive x-axis. ■

Graphical Vector Subtraction

Subtracting a vector is equivalent to adding its negative. The negative of a vector has the same magnitude but points in the opposite direction.

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

To subtract $\vec{B}$ from $\vec{A}$:

1. Draw vector $\vec{A}$.
2. Draw vector $-\vec{B}$ (same magnitude as $\vec{B}$, but opposite direction) from the head of $\vec{A}$.
3. The resultant vector is drawn from the tail of $\vec{A}$ to the head of $-\vec{B}$.

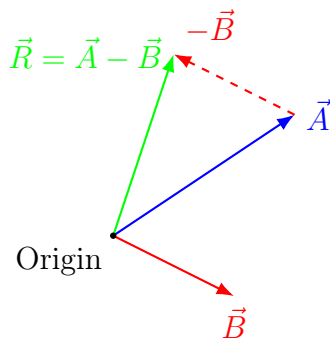


Figure 1.5: Graphical vector subtraction ($\vec{A} - \vec{B}$). Vector $-\vec{B}$ has the same magnitude as $\vec{B}$ but points in the opposite direction. The resultant is found by adding $\vec{A}$ and $-\vec{B}$ head-to-tail.

Application: Calculating the change in velocity of a moving object, which is essential for determining acceleration (a vector). If a vehicle's initial velocity is $\vec{v}_i$ and its final velocity is $\vec{v}_f$, the change in velocity is $\Delta\vec{v} = \vec{v}_f - \vec{v}_i$.

Practice Task 3

For each problem below, determine the resultant vector. You may use any of the graphical methods (Parallelogram or Polygon) or the Component Method as appropriate. Show your work clearly for each step.

1. **Force on a Component:** Two forces act on a small component on an assembly line. Force $\vec{F}_1$ is 40 N directed horizontally to the right. Force $\vec{F}_2$ is 60 N directed vertically upwards. Find the magnitude and direction of the resultant force.
2. **Architectural Surveying:** A surveyor walks 20 m East, then 15 m North, and finally 10 m at 30° South of West. What is the surveyor's total displacement (magnitude and direction) from the starting point?
3. **Water Current in a Pipe:** Water in a section of pipe has a velocity of 5 m/s directly forward. It then encounters a cross-current that imparts a velocity of 2 m/s perpendicular to the initial flow. What is the resultant velocity (magnitude and direction) of the water?
4. **Towing a Barge:** Two tugboats are pulling a large barge. Tugboat A pulls with a force of 5000 N at 0° (due East). Tugboat B pulls with a force of 7000 N at 45° North of East. Calculate the magnitude and direction of the total resultant force on the barge.
5. **Garment Design - Fabric Stretch:** A piece of fabric is stretched by a force of 10 N along its length. Simultaneously, it experiences a shearing force of 3 N perpendicular to its length. Determine the magnitude and direction of the resultant force on the fabric.

Chapter Test

- _____ 1. What is the primary aim of physics?
- A. To classify all living organisms
 - B. To describe the fundamental aspects of our universe
 - C. To study the chemical composition of matter
 - D. To predict economic trends
- _____ 2. Which of the following is a scalar quantity?
- A. Force
 - B. Velocity
 - C. Temperature
 - D. Acceleration
- _____ 3. Which of these is one of the seven SI **fundamental quantities**?
- A. Newton (N)
 - B. Watt (W)
 - C. Meter (m)
 - D. Volt (V)
- _____ 4. The quantity 'speed' is best described as being derived from which fundamental quantities?
- A. Mass and Time
 - B. Length and Time
 - C. Length and Mass
 - D. Time only
- _____ 5. To convert 1500 millimeters (mm) to meters (m), using the metric prefix trick, you should:
- A. Move the decimal 3 places to the right
 - B. Move the decimal 2 places to the left
 - C. Move the decimal 3 places to the left
 - D. Move the decimal 2 places to the right
- _____ 6. An electrical wire has a length of 50 inches. What is this length approximately in centimeters? (Given: 1 inch = 2.54 cm)
- A. 12.7 cm
 - B. 25.4 cm
 - C. 127 cm
 - D. 1270 cm
- _____ 7. In a graphical representation of a vector, what does the length of the arrow indicate?
- A. The vector's direction
 - B. The vector's starting point
 - C. The vector's magnitude
 - D. The vector's endpoint
- _____ 8. If a food product needs to be stored at -10°C , what is this temperature in Kelvin (K)?
- A. 263.15 K
 - B. 273.15 K
 - C. 283.15 K
 - D. 293.15 K

- _____ 9. Two forces, $\vec{F}_1$ and $\vec{F}_2$, act on an object. If $\vec{F}_1$ is 10 N North and $\vec{F}_2$ is 10 N South, what is the magnitude of their resultant force?
- A. 0 N B. 10 N C. 20 N D. 100 N
- _____ 10. A drone performs two successive displacements: $\vec{D}_1 = 5$ m East and $\vec{D}_2 = 5$ m North. Which graphical method would be most straightforward to find the resultant displacement by drawing?
- A. Parallelogram Method C. Polygon Method
B. Component Method D. Direct measurement (without method)
- _____ 11. Which of the following quantities is considered a **derived quantity**?
- A. Amount of Substance C. Pressure
B. Luminous Intensity D. Time
- _____ 12. An architectural drafter receives a drawing with dimensions in feet. For precise cutting, they need to convert a length of 7.5 feet to meters. Approximately how many meters is this? (Given: 1 foot = 0.3048 m)
- A. 2.29 m B. 22.86 m C. 229 m D. 0.229 m
- _____ 13. A hydraulic system exerts a force of 200 N to the right and another force of 150 N downwards. If you were to use the Component Method to find the resultant force, what would be the x-component of the resultant force?
- A. 50 N B. 150 N C. 200 N D. 350 N
- _____ 14. When performing vector subtraction, $\vec{A} - \vec{B}$, which operation is mathematically equivalent?
- A. Adding $\vec{B}$ to $\vec{A}$ C. Multiplying $\vec{A}$ by $\vec{B}$
B. Adding $-\vec{B}$ to $\vec{A}$ D. Dividing $\vec{A}$ by $\vec{B}$
- _____ 15. Consider a food technologist measuring the temperature of an oven at 200°C and another technologist measuring the force exerted by a mixer at 50 N North. Which statement correctly identifies the type of quantity for each measurement?
- A. Temperature is a vector; Force is a scalar.
B. Temperature is a scalar; Force is a vector.
C. Both Temperature and Force are scalar quantities.
D. Both Temperature and Force are vector quantities.

Chapter Summary

- **What Physics Is:** Physics is the fundamental science describing the universe's basic mechanisms, enabling understanding and innovation across diverse fields.
- **Physical Quantities and Units:**
 - **Physical Quantities** are measurable properties (e.g., length, mass, time, temperature).
 - **Units of Measurement** are standard quantities used to express physical quantities. Key systems include the **International System of Units (SI)** and the **Imperial System**.
 - **Fundamental Quantities** (Base Quantities) are independent building blocks of measurement (the seven SI fundamental quantities are Length, Mass, Time, Electric Current, Temperature, Amount of Substance, and Luminous Intensity).
 - **Derived Quantities** are combinations of fundamental quantities (e.g., Speed is derived from Length and Time; Force from Mass, Length, and Time).
- **Unit Conversion:**
 - Essential for precision and compatibility across different unit systems (e.g., SI to Imperial) and within the metric system.
 - Performed using **conversion factors** or the "**moving decimal**" **trick** for metric prefixes.
- **Introduction to Vectors:**
 - **Scalar Quantity:** Has only **magnitude** (e.g., temperature, mass, speed).
 - **Vector Quantity:** Has both **magnitude** and **direction** (e.g., displacement, velocity, force).
 - Vectors are graphically represented by **arrows**, where length indicates magnitude and orientation indicates direction. **Scaling** is used to represent magnitudes on diagrams.
 - Vectors can also be represented **algebraically** using components (e.g., $\vec{F} = F_x\hat{i} + F_y\hat{j}$).
- **Vector Operations (Finding the Resultant Vector):**
 - The **resultant vector** represents the single, equivalent effect of multiple vectors combined, crucial for analyzing complex systems like forces on structures or machine movements.
 - **Graphical Addition Methods:**

- * **Parallelogram Method:** For two vectors originating from the same point; involves drawing a parallelogram and finding the diagonal from the common origin.
- * **Polygon (Head-to-Tail) Method:** For two or more vectors; involves drawing vectors consecutively (tail-to-head), with the resultant drawn from the first tail to the last head.
- **Component Method for Addition:** A precise, analytical method involving:
 1. Resolving each vector into its x and y components ($V_x = V \cos(\theta)$, $V_y = V \sin(\theta)$).
 2. Summing all x-components and y-components separately to get R_x and R_y .
 3. Calculating the resultant magnitude: $R = \sqrt{R_x^2 + R_y^2}$.
 4. Calculating the resultant direction: $\theta_R = \arctan\left(\frac{R_y}{R_x}\right)$ (adjusting for quadrant if needed).
- **Graphical Vector Subtraction:** Equivalent to adding the negative of a vector ($\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$), where $-\vec{B}$ has the same magnitude but opposite direction of $\vec{B}$.

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